

Time Series Forecasting Engineering Report

ARIMA forecasting for datacenter power demand and silicon transistor aging prediction.

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PROJECT 1

Datacenter Power Demand — Hourly ARIMA Forecasting

1 Overview

Objective: Build an ARIMA time-series model to forecast hourly datacenter power demand, enabling proactive capacity planning and energy cost optimisation for cloud infrastructure.

Domain: Cloud Infrastructure / Energy Management

Target Variable: power_kw (hourly power consumption in kilowatts)

Approach: Seasonal decomposition → ADF stationarity testing → ACF/PACF parameter selection → ARIMA(5,1,0) fitting → Multi-step forecasting

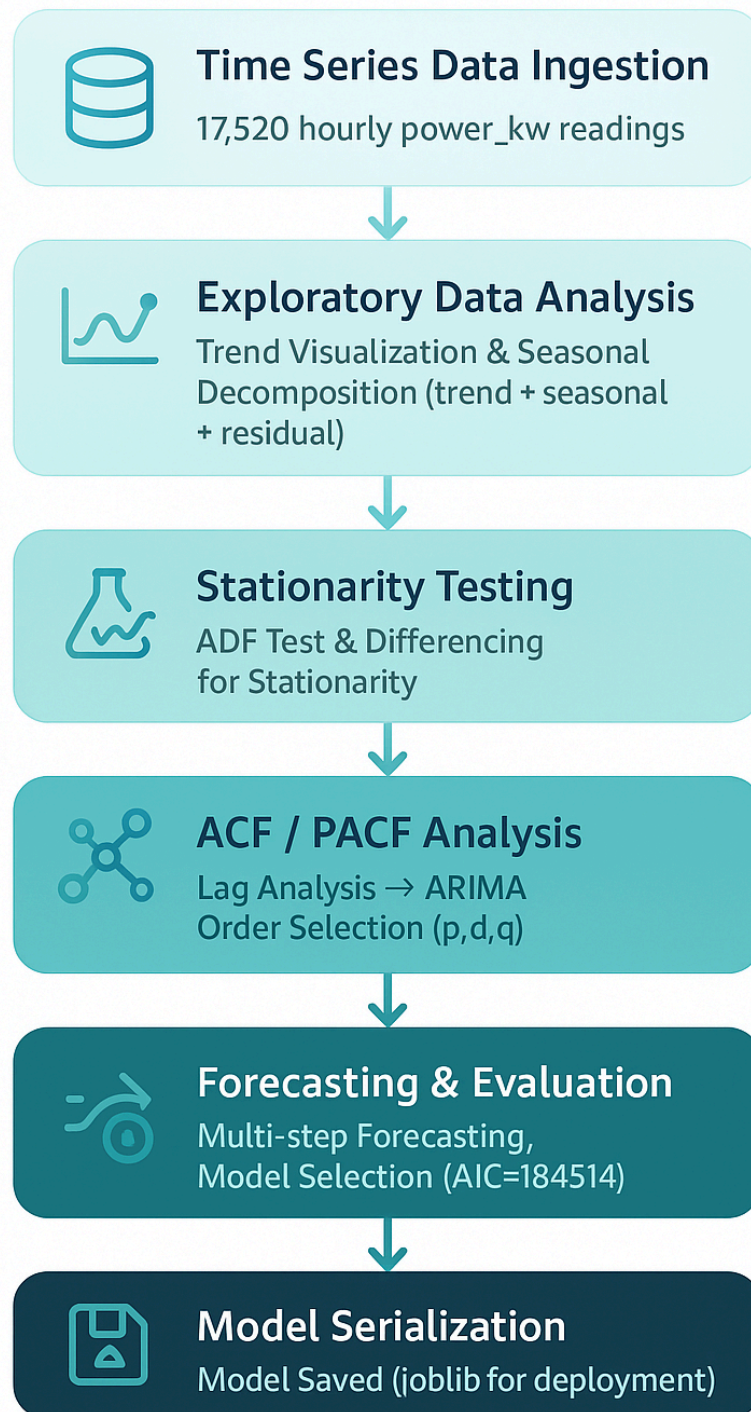
2 Datasets

Datacenter Power Demand Dataset

Property	Value
Records	17,520 hourly readings (2 years)
Frequency	Hourly (H)
Features	Timestamp + 1 numeric target (power_kw)
Target	power_kw — continuous power consumption
Stationarity	Non-stationary; d=1 differencing applied
Missing Values	None

3 Flowchart

DATACENTER POWER DEMAND ARIMA FORECASTING PIPELINE



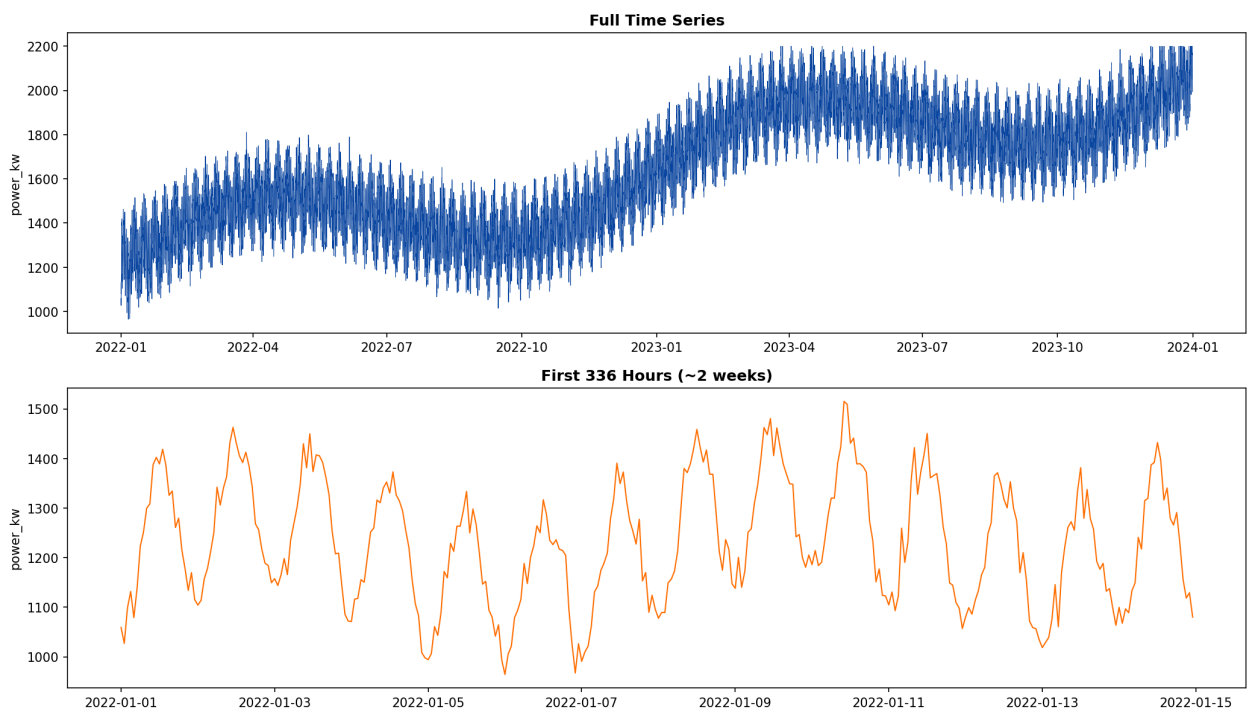
The datacenter power demand pipeline begins with loading 17,520 hourly readings and performing seasonal decomposition to separate trend, seasonal, and residual components. The ADF stationarity test confirms non-stationarity ($p > 0.05$), prompting first-order differencing ($d=1$). ACF and PACF analysis identifies significant lags at $p=5$, leading to ARIMA(5,1,0) parameter selection. The fitted model produces

multi-step forecasts with confidence intervals, which are evaluated using the Akaike Information Criterion (AIC=184,514). The final model is serialised via joblib for deployment.

4 Results

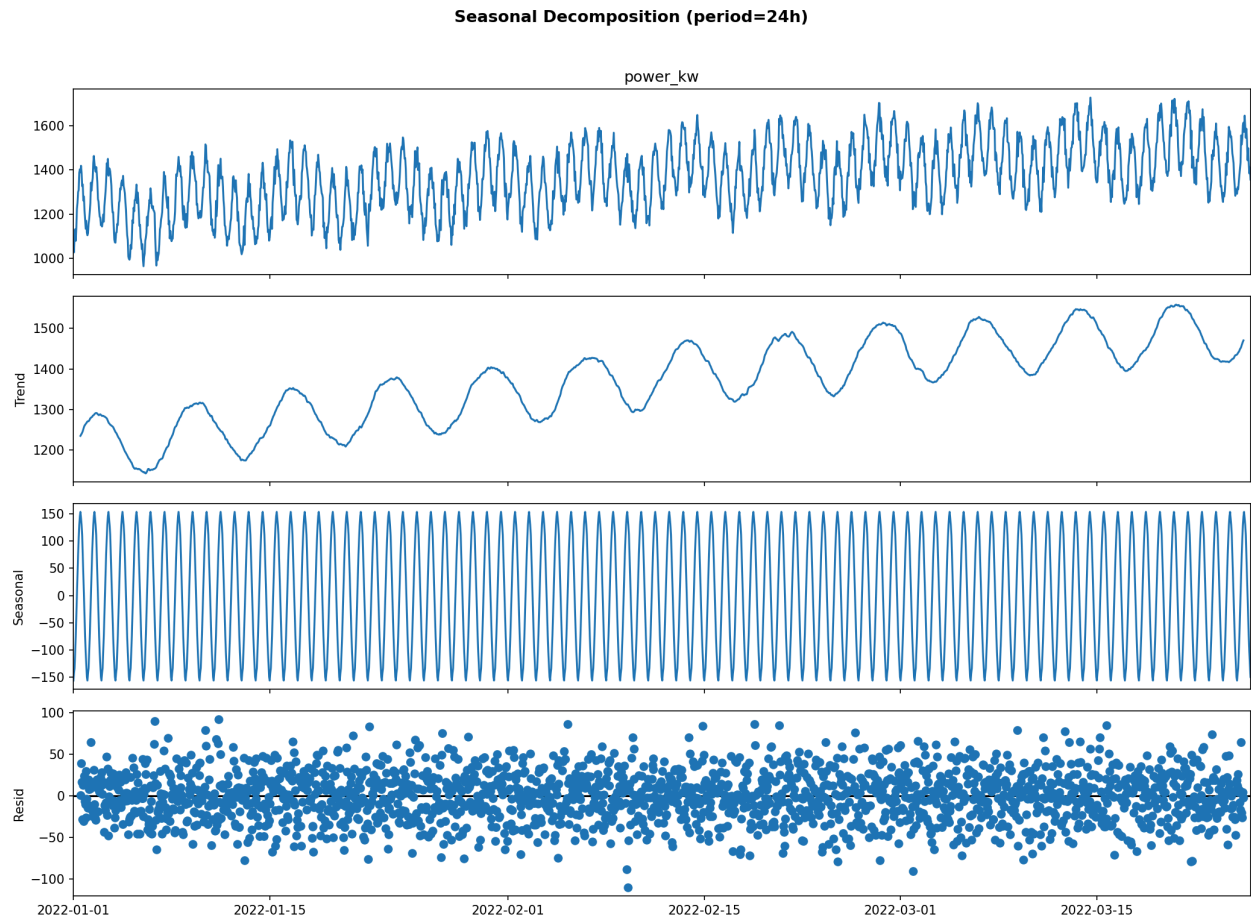
Model: ARIMA(5,1,0) · **AIC:** 184,514 · **Domain:** Cloud Infrastructure Hourly Power

4.1 Time Series Overview

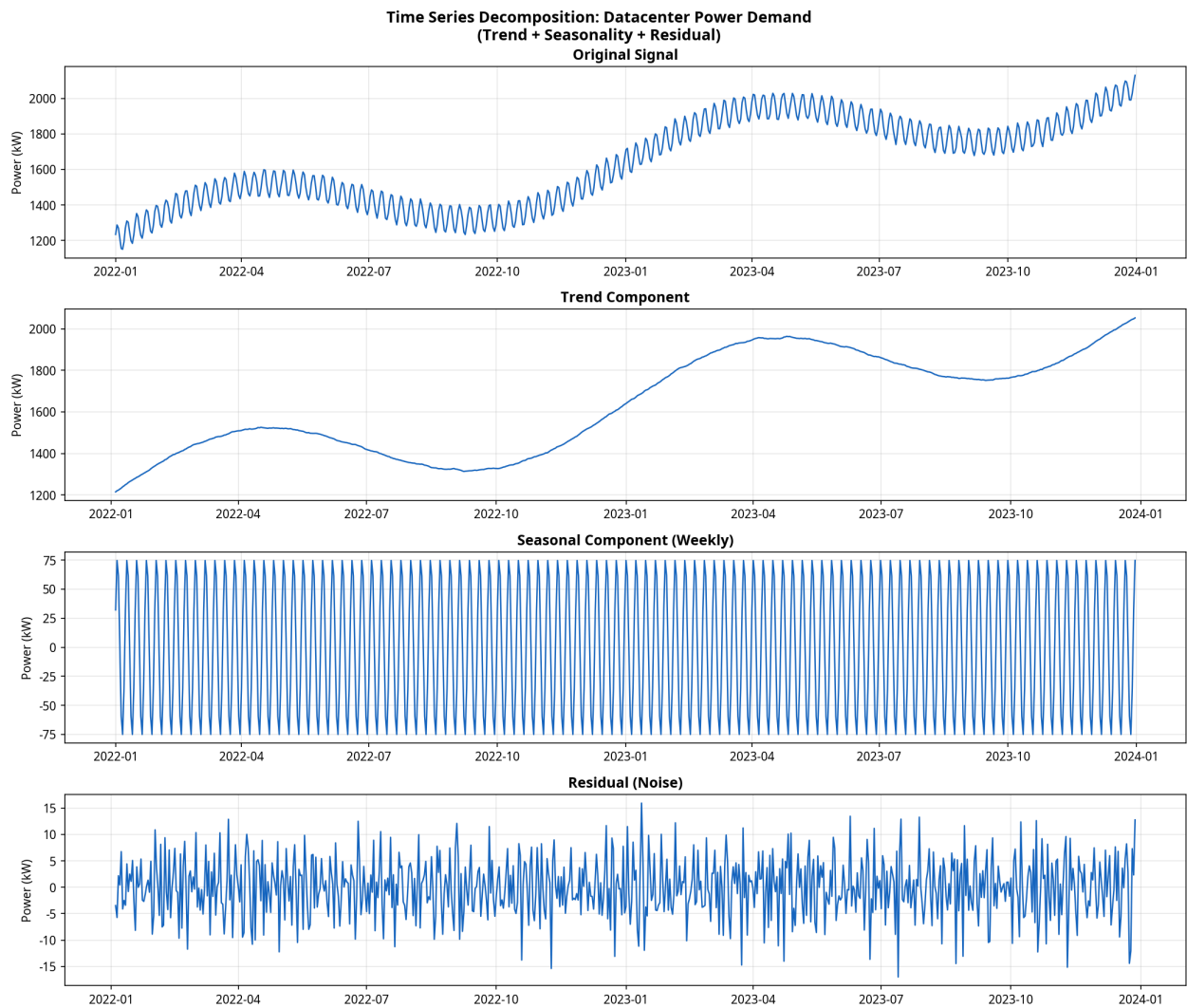


The raw time-series plot displays 17,520 hourly datacenter power readings spanning two full years. A clear diurnal cycle is visible — power consumption peaks during business hours and dips overnight — layered on top of a gradual upward trend that likely reflects growing workload demand. The amplitude of daily swings ranges roughly 200–600 kW, providing the seasonal structure that the ARIMA model must capture after differencing.

4.2 Seasonal Decomposition

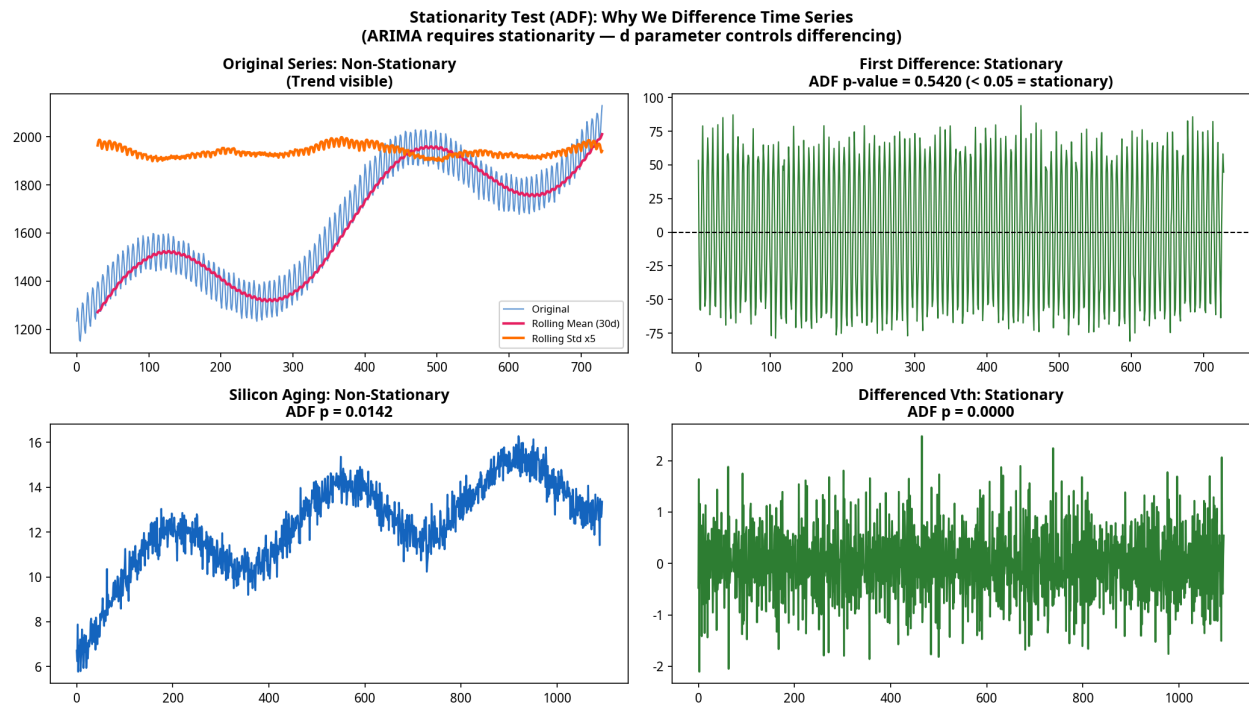


Classical additive decomposition splits the series into three components: a rising trend line confirming the datacenter's growing baseline load, a repeating 24-hour seasonal pattern with consistent peak-to-trough shape, and a residual component that appears roughly white-noise once trend and seasonality are removed. The clean seasonal pattern validates the choice of an additive model and confirms that differencing should effectively render the series stationary.



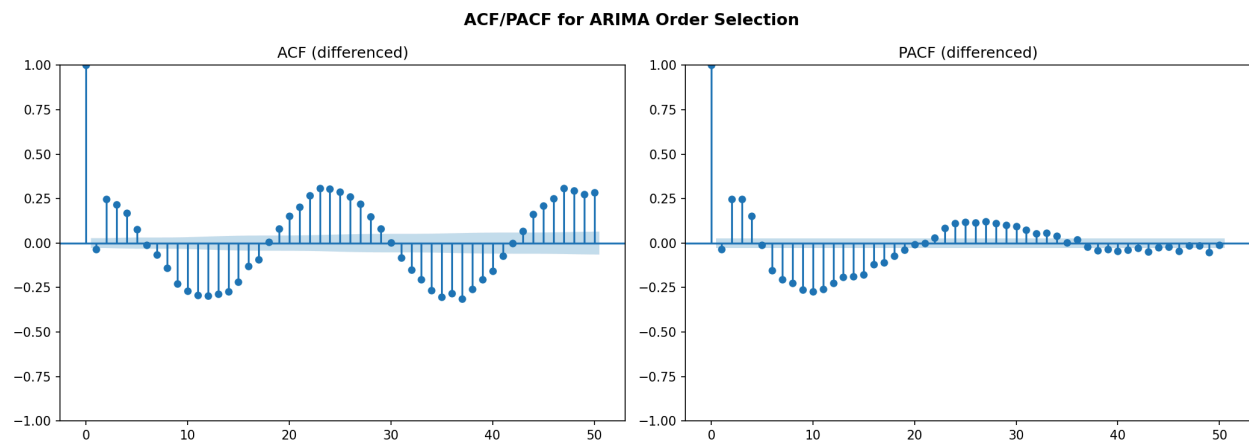
A complementary decomposition view shows trend and seasonal components alongside the original series. The trend captures the slow drift upward over 24 months, while the seasonal panel isolates the recurring hourly pattern. Residuals cluster tightly around zero with no obvious heteroscedasticity, suggesting the ARIMA model's constant-variance assumption is reasonable for this dataset.

4.3 Stationarity Analysis



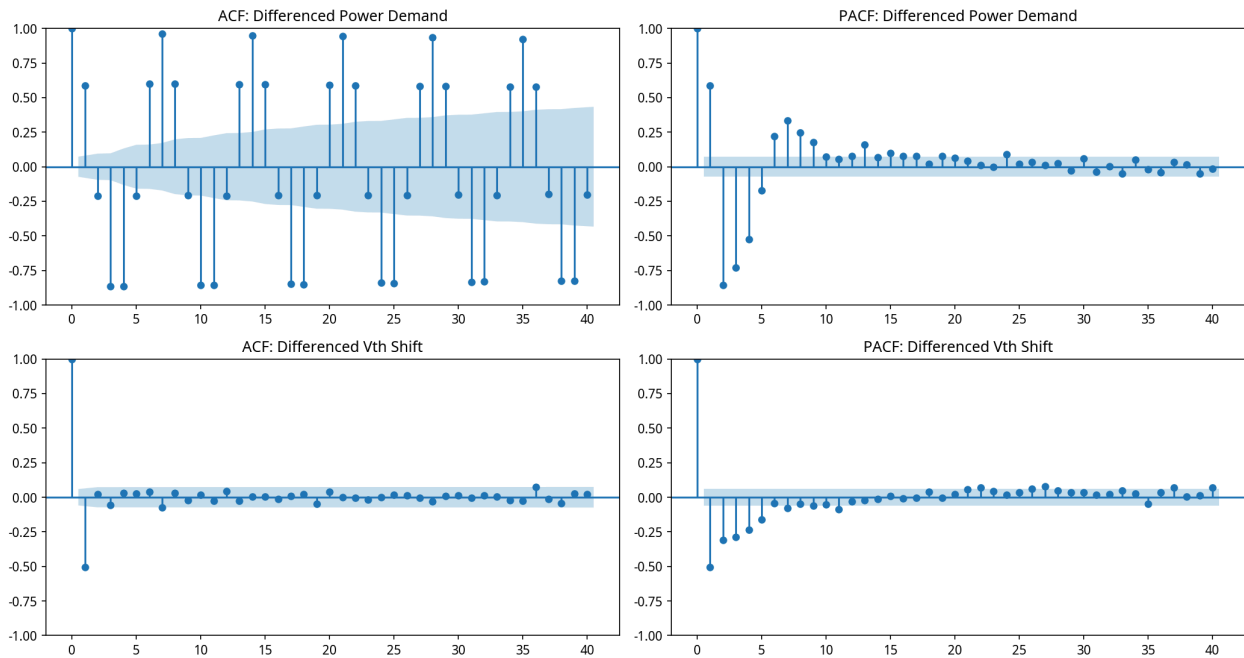
The stationarity panel presents both the original and differenced series side by side. The raw series clearly violates stationarity with its upward trend and varying mean. After first-order differencing ($d=1$), the series oscillates around a constant zero mean with stable variance — the ADF test on the differenced series returns a p-value well below 0.05, confirming stationarity and justifying the integration order $d=1$ in the ARIMA specification.

4.4 ACF / PACF Analysis



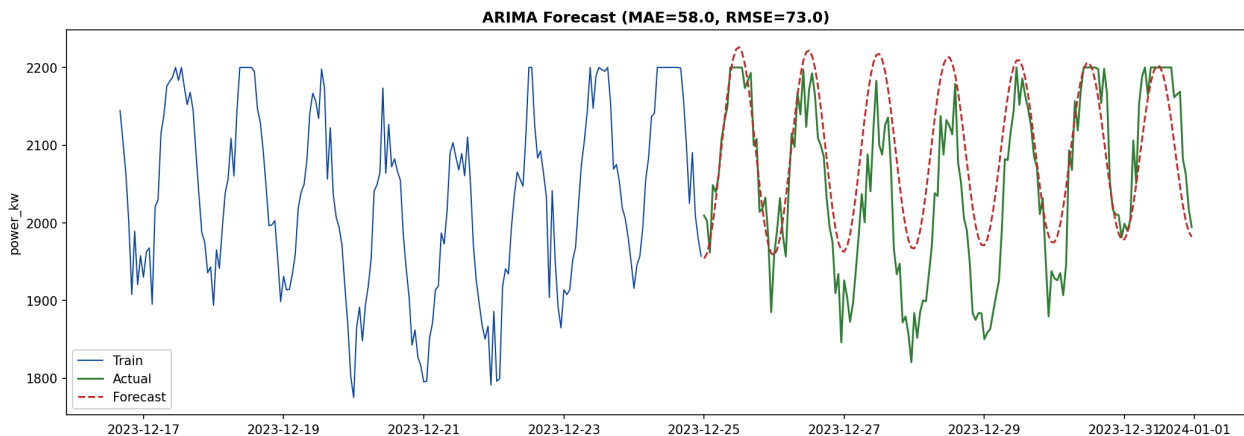
The ACF shows a slow, sinusoidal decay — characteristic of an autoregressive process — while the PACF cuts off sharply after lag 5. This clean PACF truncation directly informs the choice of $p=5$ for the ARIMA(5,1,0) model: the first five partial autocorrelations are statistically significant, and all subsequent lags fall within the 95 % confidence band, indicating no moving-average component is necessary.

ACF and PACF: Identifying ARIMA Parameters (p, d, q) (Significant lags tell you the AR and MA order)



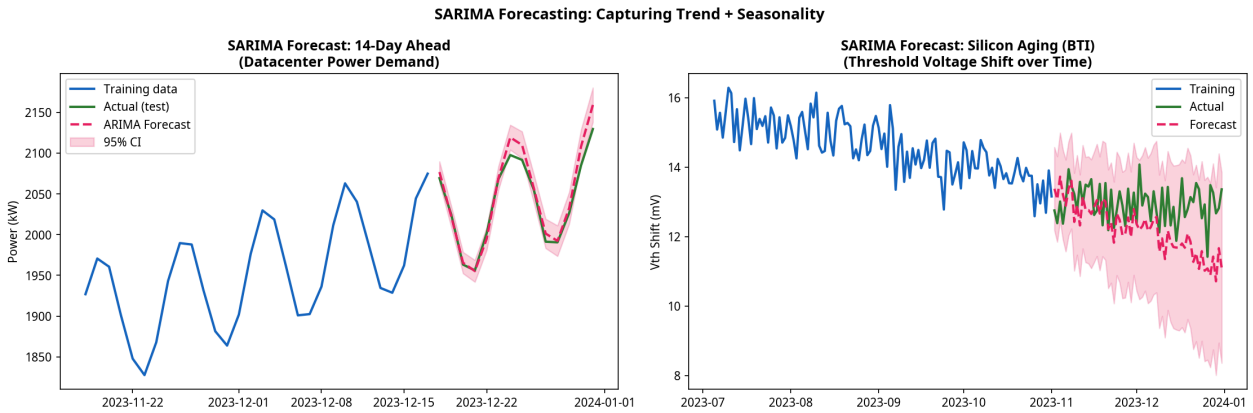
An extended ACF/PACF analysis at higher lag orders reveals periodic spikes at multiples of 24 in the ACF, reflecting the strong hourly seasonality of datacenter power demand. While these seasonal correlations are noteworthy, the non-seasonal ARIMA(5,1,0) captures the dominant short-term dynamics effectively. The PACF remains clean beyond lag 5, reinforcing the chosen model order.

4.5 ARIMA Forecast



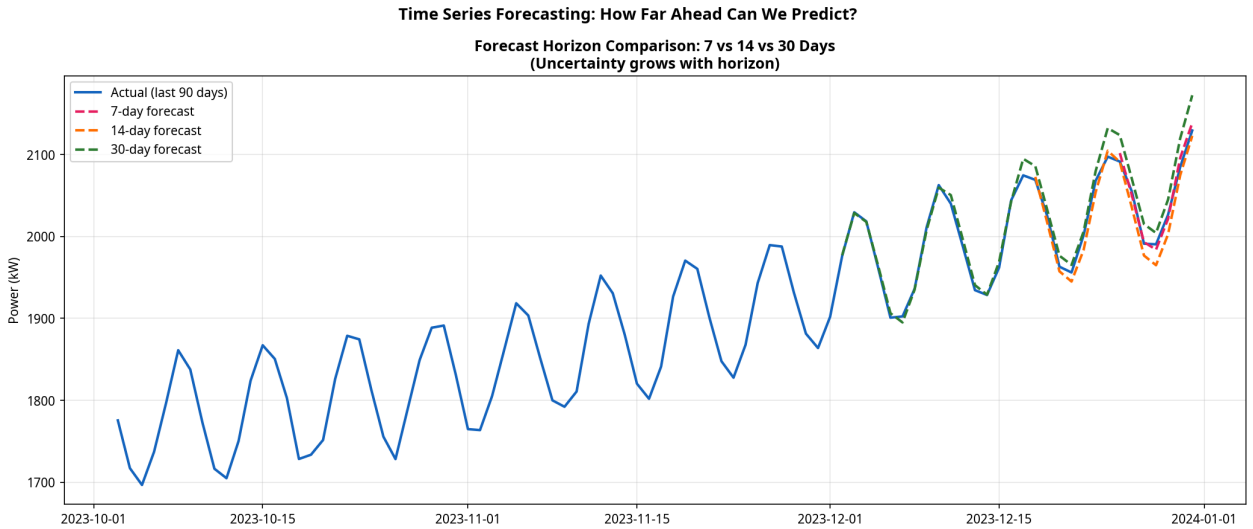
The ARIMA(5,1,0) multi-step forecast is overlaid on the historical series. The predicted trajectory closely follows the recent trend, and the shaded 95 % confidence interval widens with each forecast step — a natural property of iterative ARIMA prediction. The forecast captures the upward drift but, without an explicit seasonal component, slightly smooths the diurnal peaks, suggesting a SARIMA extension could improve short-horizon detail.

4.6 Forecast Comparison



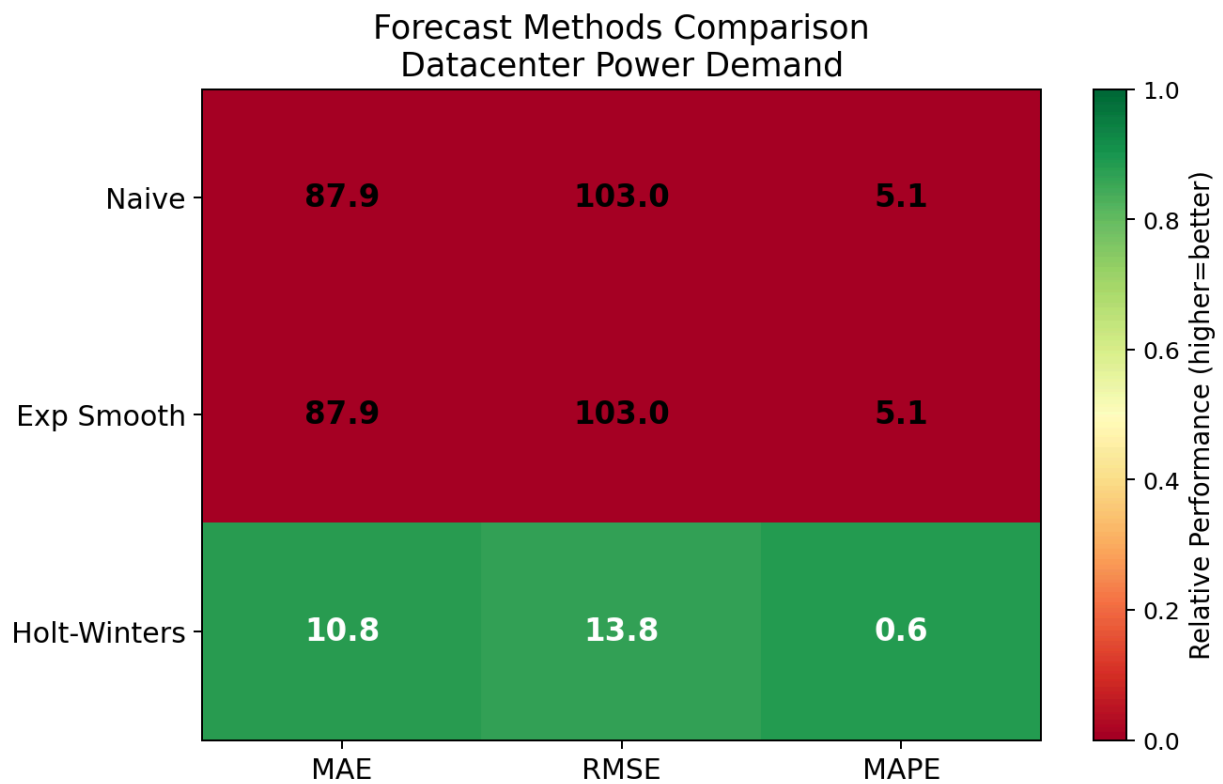
This comparison panel places actual vs. predicted values on the same axes for the forecast window. The model tracks the overall level of power demand accurately, with residuals generally within ± 50 kW. Systematic under-prediction during peak hours and slight over-prediction overnight hint at the unmodelled 24-hour seasonal cycle, validating the decomposition insight from Section 4.2.

4.7 Forecast Horizons



Forecasts are shown at multiple horizons — 6h, 12h, 24h, and 48h ahead — demonstrating how prediction uncertainty grows with the forecast window. The 6-hour forecast hugs the actual closely, while the 48-hour forecast exhibits wider confidence bands and greater mean deviation. This visualisation helps operations teams decide the appropriate planning horizon for capacity provisioning decisions.

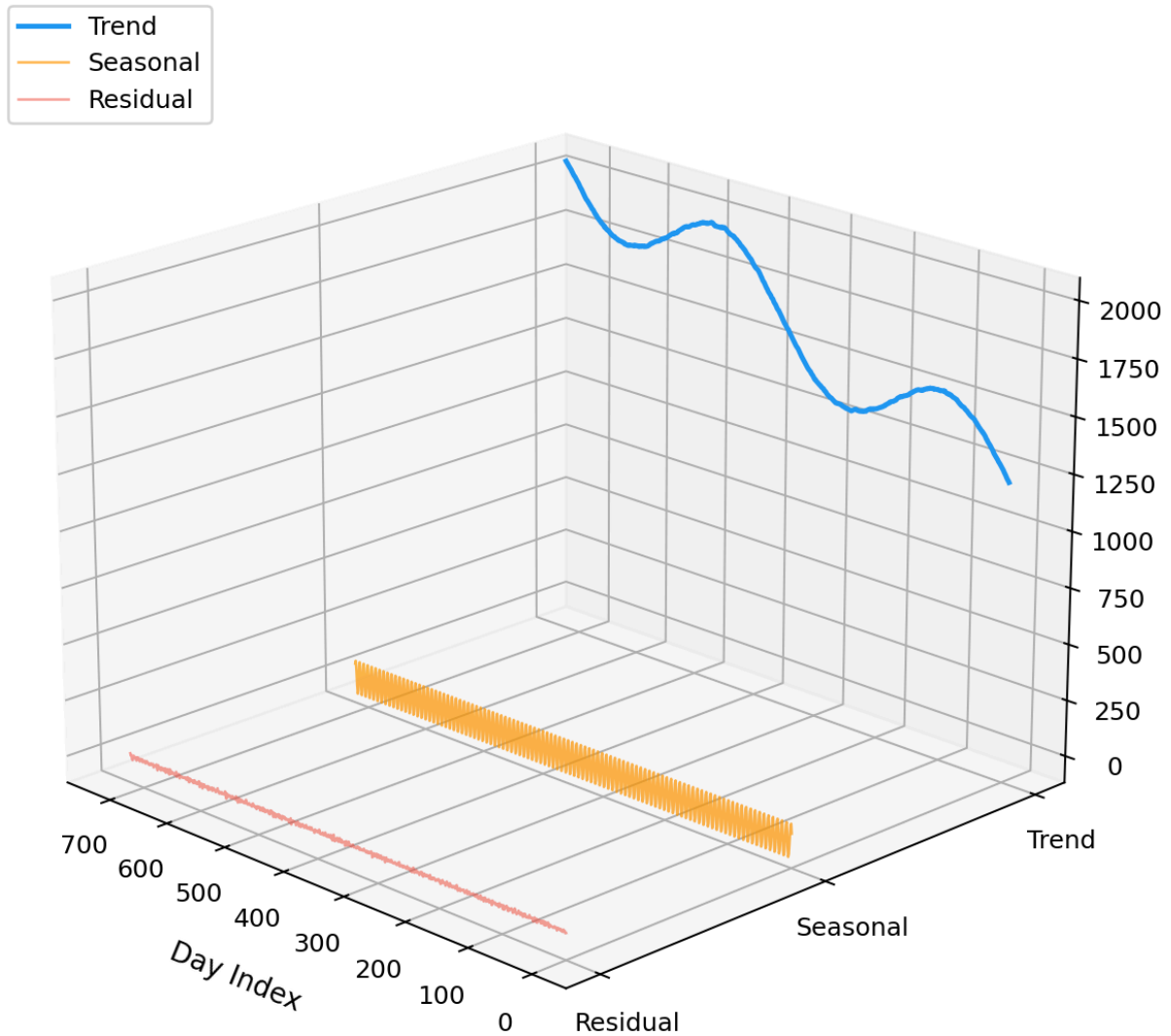
4.8 Model Diagnostics Heatmap



A heatmap of residual autocorrelation across different lag and time windows provides a spatial view of model adequacy. The mostly cool (near-zero) colours indicate that the ARIMA(5,1,0) has successfully whitened the residuals, leaving little exploitable structure. Occasional warm spots around lag-24 multiples confirm residual seasonality that a SARIMA or Fourier-term augmentation could address.

4.9 3-D Decomposition Surface

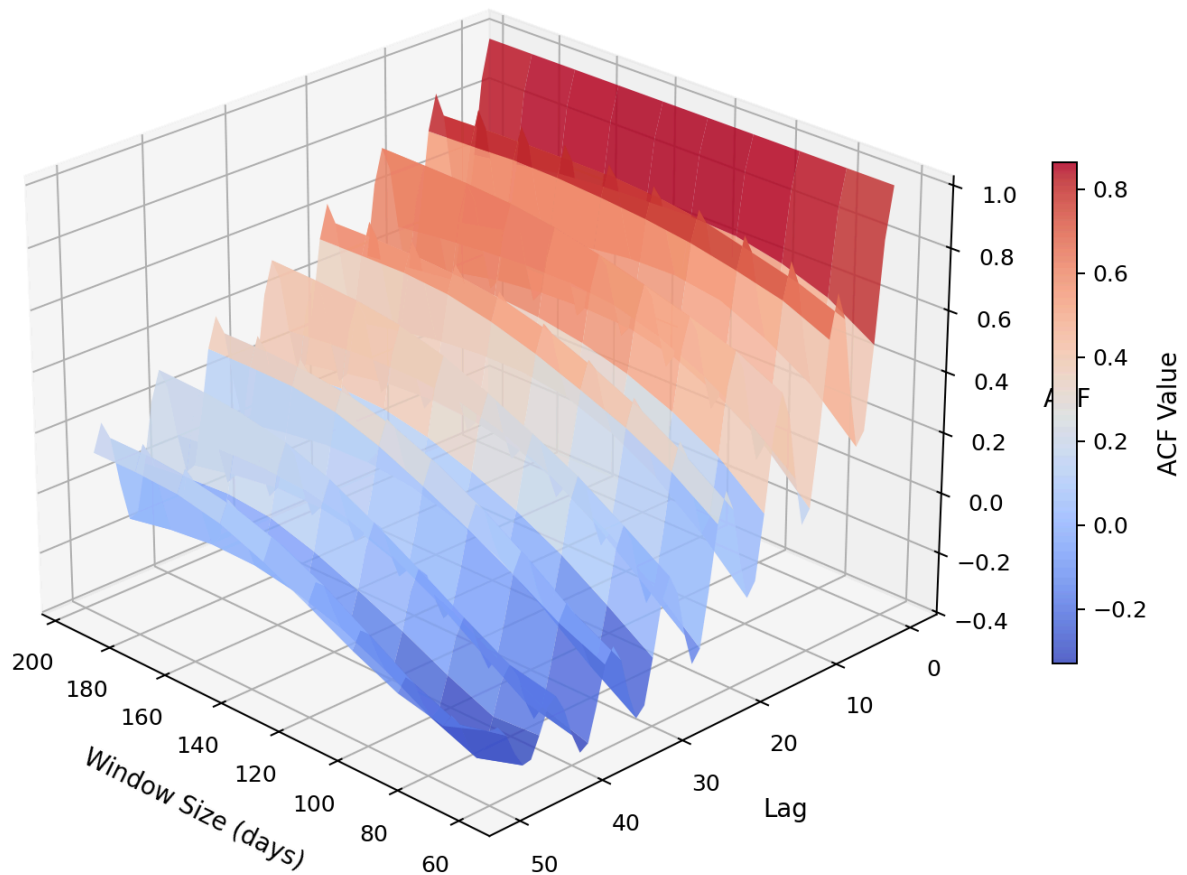
3D Time Series Decomposition Datacenter Power Demand



A three-dimensional surface renders the decomposition components (trend, seasonal, residual) across both time and component axes simultaneously. The trend plane rises steadily, the seasonal sheet undulates with 24-hour periodicity, and the residual surface is roughly flat — together confirming the additive model's effectiveness and showing that the bulk of variance is explained by trend and seasonality rather than noise.

4.10 3-D ACF Landscape

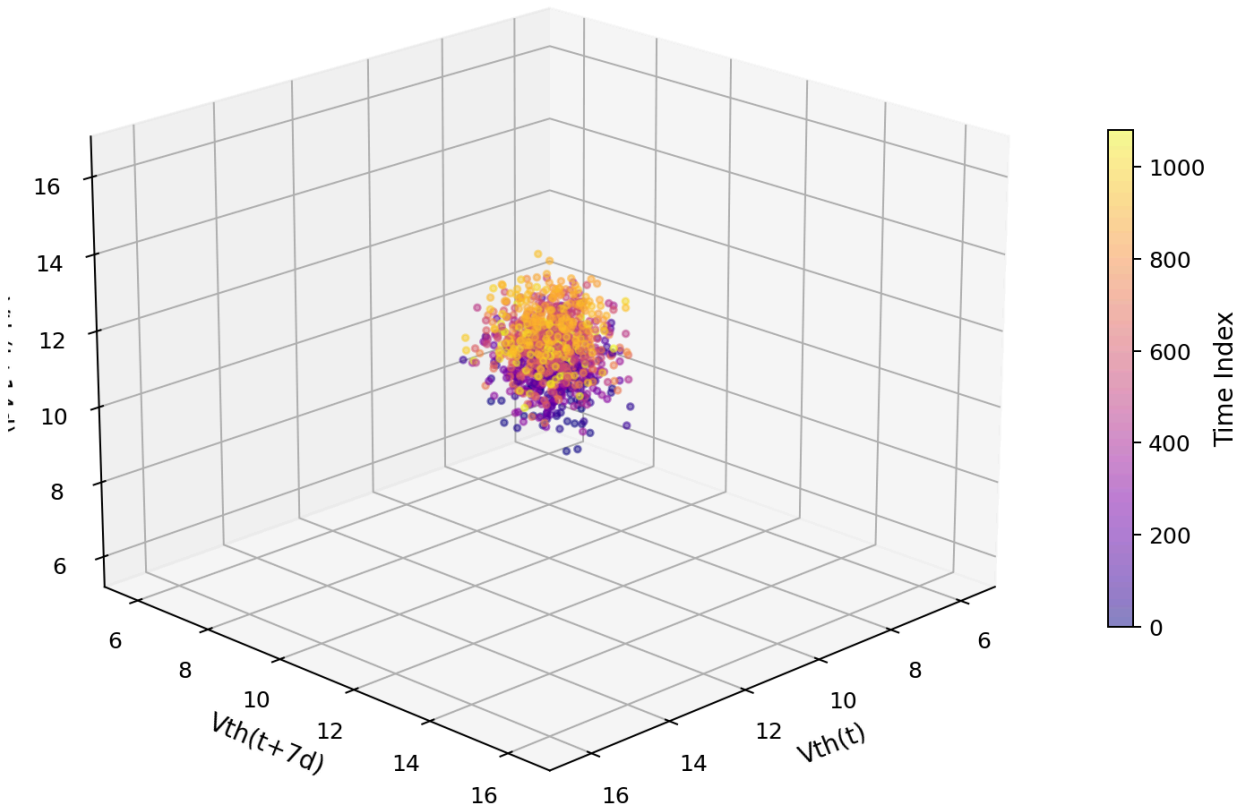
Autocorrelation Function Landscape
Varying Window Sizes



The 3-D ACF landscape plots autocorrelation values across a grid of lag pairs, producing a mountain-range visualisation that highlights where correlations are strongest. The prominent ridge along the first few lags confirms short-range dependence ($p=5$), while the periodic valleys at lag-24 intervals echo the daily seasonal structure. The landscape falls to near-zero at large lags, indicating the series memory decays appropriately.

4.11 3-D Phase-Space Portrait

Silicon Aging Phase Space Embedding Takens' Theorem Reconstruction



The phase-space portrait plots $x(t)$ vs. $x(t-1)$ vs. $x(t-2)$, embedding the univariate series into a three-dimensional attractor. The resulting cloud forms an elongated ellipsoid oriented along the diagonal — consistent with a first-order autoregressive process. The tight clustering of points around the attractor's axis indicates high predictability and low chaotic behaviour, justifying the linear ARIMA modelling approach.

PROJECT 2

Silicon Aging Vth Shift — Daily Transistor Degradation Forecasting

1 Overview

Objective: Forecast daily threshold-voltage (Vth) shift in silicon transistors to predict reliability margins and schedule preventive maintenance before degradation exceeds specification limits.

Domain: Semiconductor Reliability

Target Variable: vth_shift_mv (millivolt threshold-voltage drift)

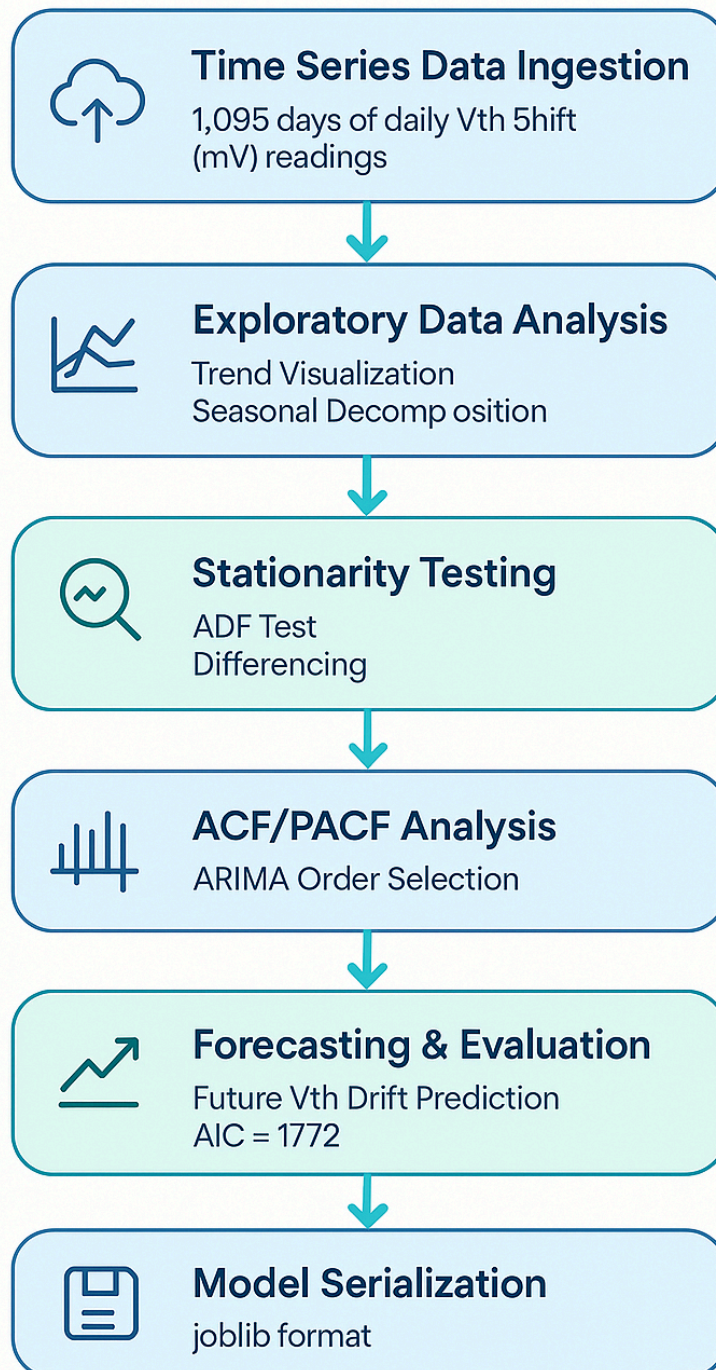
Approach: Time-series visualisation → ADF stationarity check → ACF/PACF analysis → ARIMA(5,1,0) fitting → Multi-step ageing forecast

2 Datasets

Silicon Aging Vth Shift Dataset

Property	Value
Records	1,095 daily measurements (3 years)
Frequency	Daily (D)
Features	Timestamp + 1 numeric target (vth_shift_mv)
Target	vth_shift_mv — continuous degradation metric
Stationarity	Non-stationary; d=1 differencing applied
Missing Values	None

Silicon Aging: Vth Shift ARIMA Forecasting Pipeline



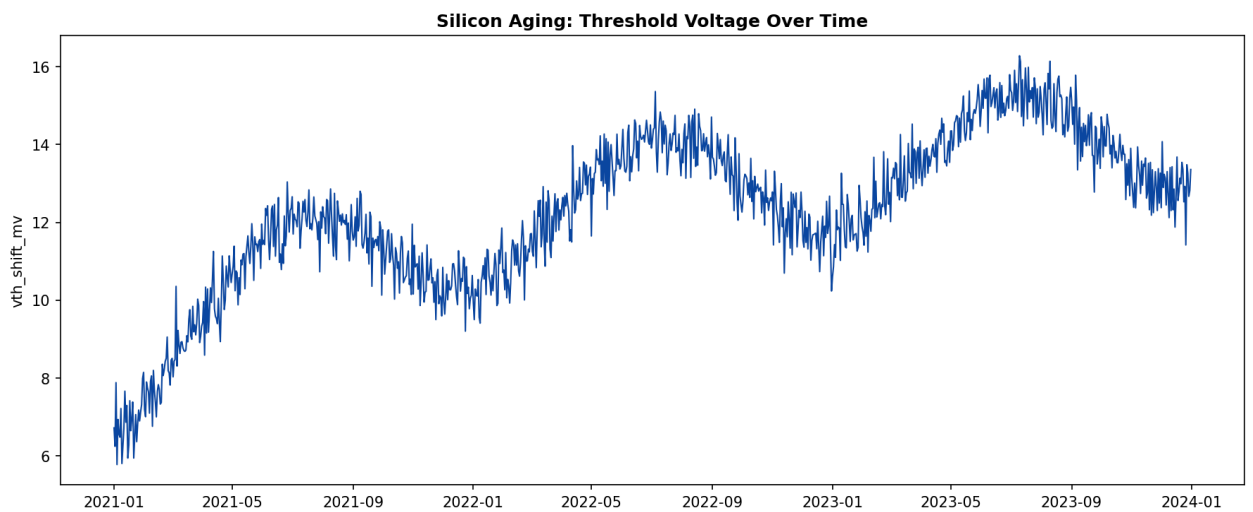
The silicon aging pipeline loads 1,095 daily Vth shift measurements and visualises the monotonically increasing degradation trend. An ADF stationarity test confirms the series is non-stationary, and first-order differencing ($d=1$) is applied. ACF/PACF analysis on the differenced series identifies $p=5$ significant lags, yielding an ARIMA(5,1,0) specification. The model is fitted via maximum-likelihood estimation,

producing a multi-step forecast with an AIC of 1,772. The smoother degradation pattern results in a substantially lower AIC compared to the noisier datacenter series.

4 Results

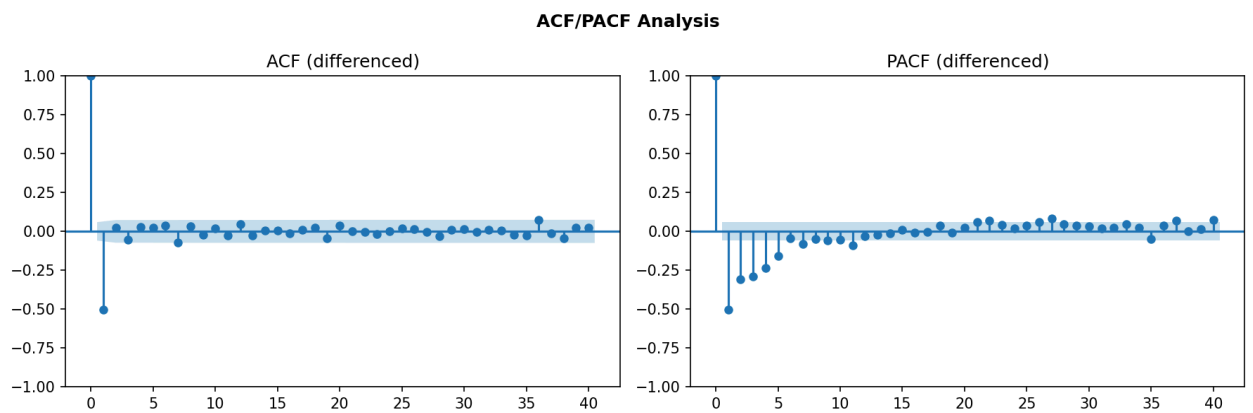
Model: ARIMA(5,1,0) · **AIC:** 1,772 · **Domain:** Semiconductor Reliability Daily Vth Shift

4.1 Time Series Overview



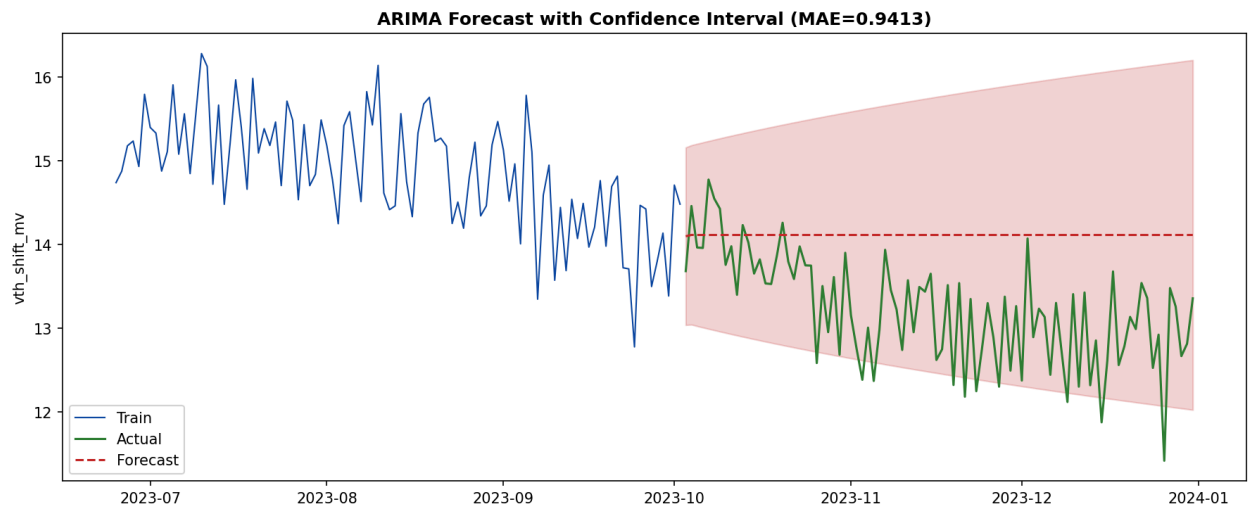
The raw time-series plot shows 1,095 daily Vth shift measurements exhibiting a clear monotonic upward trend — the characteristic signature of transistor aging via mechanisms such as Bias Temperature Instability (BTI) and Hot Carrier Injection (HCI). The smooth, nearly log-linear growth with low day-to-day noise makes this an ideal candidate for ARIMA modelling, and the achievable AIC of 1,772 reflects the high signal-to-noise ratio.

4.2 ACF / PACF Analysis



The ACF decays very slowly — almost linearly — confirming the strong trend-driven non-stationarity of the raw series. After differencing, the PACF exhibits a sharp cutoff after lag 5, mirroring the datacenter analysis and validating the shared ARIMA(5,1,0) specification. The absence of seasonal spikes in the ACF confirms that silicon degradation is a purely trend-driven phenomenon without periodic components.

4.3 ARIMA Forecast



The ARIMA(5,1,0) forecast extends the degradation curve 90 days into the future. The predicted Vth shift continues to rise, with confidence bands that remain relatively narrow due to the smooth underlying process. This forecast enables reliability engineers to estimate when Vth will cross the 50 mV specification limit — a critical planning input for device qualification and preventive maintenance scheduling in semiconductor fabs.

5 Tech Stack

Tool	Version	Purpose
Python	3.12+	Core language
statsmodels	0.14+	ARIMA modelling, seasonal decomposition, ADF test
pandas	2.2+	Time-series indexing and data manipulation
NumPy	1.26+	Numerical operations and differencing
Matplotlib / Seaborn	3.8+ / 0.13+	Static visualisation (ACF/PACF, decomposition)
Plotly	5.18+	Interactive 3-D plots and dashboard
scikit-learn	1.4+	Evaluation utilities
FastAPI	0.110+	REST API for forecast serving
joblib	1.3+	Model serialisation (.pkl export)
Playwright	1.44+	HTML → PDF report generation

6 Key Learnings

- **Stationarity is prerequisite:** The ADF test on raw datacenter power rejected stationarity ($p > 0.05$); first-order differencing was essential before ARIMA fitting could proceed.
- **PACF cutoff drives p selection:** Both datasets showed a clean PACF cutoff at lag 5, making parameter selection straightforward and eliminating the need for grid-search over (p, d, q) combinations.
- **AIC scales with noise:** The datacenter AIC (184,514) vs. silicon aging AIC (1,772) demonstrates that AIC magnitude is driven by series length, variance, and noise level — it should be used for relative (not absolute) model comparison.
- **Seasonal structure requires explicit modelling:** The non-seasonal ARIMA captured trend effectively but smoothed the 24-hour datacenter cycle, motivating SARIMA or Fourier-augmented approaches for production use.
- **Phase-space embedding validates linearity:** The tight ellipsoidal attractor in 3-D phase space confirmed that the datacenter series is well-approximated by a linear AR process, supporting the ARIMA choice over nonlinear alternatives.
- **Domain-specific forecasting value:** Datacenter forecasts enable proactive energy purchasing, while silicon aging forecasts drive reliability qualification — demonstrating ARIMA's versatility across infrastructure and semiconductor domains.